MIL-G-18853C(OS)
29 August 1975
Superseding
MIL-G-18853B(Wep)
5 February 1962 with
Amendment - 7
3 June 1969

MILITARY SPECIFICATION

GYRO, RATE

(FOR TORPEDO MARK 37 MODS 2 AND 3)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification describes the requirements for the procurement, inspection, and test of a rate gyro.

2. APPLICABLE DOCUMENTS

2.1 Referenced documents. The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation - Packaging, Methods of
MIL-E-82590	Environmental Requirements and Tests (For Torpedo MK 37 Components)

STANDARDS

Military

MIL-STD-454	Standard General Requirements for Electronic Equipment

DRAWINGS

Naval Sea Systems Command (Code Ident. 10001)

LD 480480	Rate Gyro Assembled, and all parts and subassemblies listed thereon

FSC 1355

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LD 495855	Shipping (Packing) Condition
Dwg 859607	Rate Gyro Schematic Diagram
Dwg 1503477	Rate Gyro Assembled
Dwg 1523749	Cover
Dwg 1539007	Setscrew, Hex, Socket Drive
Dwg 1874064	Gyro Fixture

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other documents. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

National Bureau of Standards Publication

Handbook H28	Screw-Thread Standards for Federal Services

(Applications for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

National Motor Freight Traffic Association, Inc. , Agent

National Motor Freight" Classification

(Application for copies should be addressed to American Trucking Associations, ATTN: Traffic Order Section, 1616 P Street, N.W. Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Applications for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

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3. REQUIREMENTS

3.1 Description. The rate gyro is a device which utilizes gyroscopic action to provide a means for measuring angular turning rate and providing a corresponding electrical signal. The device comprises an electrically powered rotor that provides rotational momentum, and is mounted in a gimbal suspension. The gimbal suspension axis is disposed at a 90-degree angle with relation to the rotor spinning axis. Restraint of the gimbal is provided by means of extension springs attached between the gimbal and the housing. Whenever the rate gyro is subjected to an angular velocity about an axis that is mutually perpendicular to both the gimbal and the rotor spinning axis, the gimbal will be subjected to a torque about the gimbal axis. This torque is proportional to the aforementioned angular velocity and causes an angular deflection about the gimbal axis of a magnitude proportional to the same aforementioned angular velocity. The direction of the torque depends on the sense, clockwise or counter-clockwise, of the angular velocity. The angular deflection is converted into an electrical signal by the electromagnetic action of a pick-off, commonly called a microsyn.

3.2 General requirements.

3.2.1 Preproduction sample. Unless otherwise specified in the contract or order, preproduction samples of the rate gyro shall be manufactured using the methods and procedures proposed for the production lot . These units shall be tested as specified in Section 4 herein and are for the purpose of determining, prior to starting production, that the contractor's production methods are capable of producing rate gyros that comply with the technical requirements of the contract. See 4.3.1.

3.2.2 Materials. Unless otherwise specified in the contract or order, all materials and purchased parts entering into the final assembly of the rate gyro shall conform strictly to the detailed requirements indicated on the drawings and specifications listed in LD 480480.

3.2.3 Finishes, coatings, construction, and dimensions. The finishes, coatings, construction, and dimensions for the rate gyro shall be strictly in accordance with applicable drawings, specifications, and publications listed in LD 480480.

3.2.4 Fungus and corrosion. The rate gyro shall be rendered fungus and corrosion resistant in accordance with Requirements 4 and 15 of MIL-STD-454.

3.2.5 Marking. The marking of assemblies, subassemblies, and parts shall be in accordance with the applicable drawings listed in LD 480480.

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3.2.6 Threads. Unless otherwise specified, all threads shall be in accordance with National Bureau of Standards Handbook H28. The class or fit for threads shall be as specified on the applicable drawings.

3.2.7 Dimensions. The maximum physical dimensions of the rate gyro shall conform to Dwg 1503477.

3.2.8 Wiring. The rate gyro shall be wired in accordance with Dwgs 859607 and 1503477.

3.2.9 Conflicting requirements. In the event of conflict between this specification and other documents referenced herein, the requirements of this specification shall apply.

3.3 Performance requirements and product characteristics. The rate gyro shall have the product characteristics and meet the performance requirements given below:

Note: All line, plug, and connector designations referred to in this section are shown on Dwg. 1503477.

3.3.1 Angular rates.

3.3.1.1 The rate gyro described herein shall be capable of measuring angular rates from 0 degrees per second to 33 degrees per second with a tolerance of plus 11 minus 0 degrees per second.

3.3.1.2 The rate gyro shall be capable of measuring the angular rates described in 3.3.1.1 in a clockwise and a counterclockwise direction when rotated about an axis that is mutually perpendicular to the spin axis and the gimbal axis.

3.3.2 Sensitivity. The rate gyro sensitivity shall be 0.21 ± 0.02 volt per degree per second when feeding a 25,000 ± 250 ohm resistive load, and with a 55 ± 1 volt, 400-Hertz(Hz) microsyn input. The distortion of the input voltage to the microsyn shall not exceed 1.0 percent.

Note: This sensitivity requirement applies to angular rotations as described in 3.3.1.2.

3.3.3 Maximum output. The rate gyro maximum output, when feeding a 25,000 ± 250 ohm resistive load, and with a 55 ± 1 volt microsyn input, shall be 8 ± 0.25 volts.

3.3.4 Phasing. The output voltage on terminals F and H of P601 shall be in phase with the supply voltage (terminals D and E, respectively) when the following conditions exist:

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a. The rate gyro is positioned on a turntable (by the application of 20 ± 2 inch pounds torque to each of its 3 mounting screws) so that the mounting base is at right angles to the turntable.

b. The connector plug is at the left when viewing the gyro mounting base, and

c. The turntable is rotating in a clockwise direction when viewed from above.

3.3.5 Zeroing.

3.3.5.1 Null voltage. When feeding a 25,000 ± 250 ohm resistive load and with a 55 ± 1 volt microsyn input and 115-volt three-phase 400-Hz motor input, the output voltage shall be less than 0,15 volt with the rate gyro stationary and shall be repeatable to within ± 0.03 volt when moved from the null position and allowed to return.

3.3.5.2 Motor assembly balance. With the motor running and with the rate gyro resting on either end or either side, the output voltage shall be within 0.03 volt of the null voltage obtained in 3.3.5.1.

3.3.6 Spin axis alignment. The spin axis shall be perpendicular, within 1.25 degrees, to a plane passing through the surface of the three mounting bosses on the base.

3.3.7 Linearity. The rate gyro output must be linear, with respect to the rate of change of angular velocity, to within ± 10 percent.

3.3.8 Insulation.

3.3.8.1 The insulation resistance, between all pins of P601 connected in parallel, and ground, shall be 50 megohms minimum.

3.3.8.2 All wiring shall withstand a 500-volt, 60-Hz potential for one minute, between all pins of P601 connected in parallel and ground, with no evidence of insulation breakdown.

3.3.9 Frequency. The undamped natural frequency about the gimbal axis shall be 15 Hz with a tolerance of plus 10 minus O Hz.

3.3.10 Damping. The damping about the gimbal axis shall be 3 percent of critical minimum. Three percent critical minimum represents at least 17 percent decay in amplitude of oscillation per cycle.

3.3.11 Response time. The rate gyro motor shall reach 90 percent of full speed in 45 seconds maximum, with the power supply as specified in 3.5.1.

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3.4 Operating life. The operating life of the rate gyro shall be 1000 cycles minimum at 10 minutes per cycle when operated as specified in 3,5.1 and 3.5.4.

3.5 Power requirements.

3.5.1 The rate gyro motor shall operate from a 115 ±\t 5.75 volt, 400 ± 8 Hz, single-phase power source.

3.5.2 The rate gyro motor maximum starting power shall be 25 watts.

3.5.3 The rate gyro running power shall be 5 watts maximum.

3.5.4 The rate gyro microsyn shall operate from a 55 ± 2.75 volt, 400 ± 8 Hz, single-phase power source.

3.5.5 The microsyn input current shall be 21 ± 3.15 milliamperes (ma).

3.6 Environmental requirements. The rate gyro shall meet the other applicable requirements of this specification during or after the tests defined or specified in Section 4.

3.6.1 Vibration. The rate gyro shall meet the following vibration requirements:

3.6.1.1 Vibration, nonoperating. The rate gyro shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.6.1.1.

3.6.1.2 Vibration, operating. The rate gyro shall show no damage and shall meet all the performance requirements and product characteristics of this specification during and after the test specified in 4.6.1.2.

3.6.2 High temperature and humidity, nonoperating. The rate gyro, less pack and packaging, shall show no damage and shall meet all the performance requirements and product characteristics of this specification after being tested as in 4.6.2.

3.7 Cleanup. Prior to and after final assembly, all parts, components, and the assembly shall be thoroughly cleaned of loose, spattered, or excess solder, metal chips, and other foreign matter. All burrs, sharp edges, and rosin flash shall be removed.

3.8 Workmanship. The rate gyro, including all parts and accessories, shall be constructed and finished in a manner to assure compliance with all the requirements of this specification. Particular attention shall be paid to neatness and thoroughness of soldering, wiring, impregnation of coils, marking of parts and assemblies, plating, painting, machine-screw assemblage, and freedom of parts from all burrs and sharp edges. The standards of workmanship exhibited in any approved

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preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the rate gyro shall be classified as follows:

a. Quality conformance inspection (see 4.5)

b. Preproduction inspection and periodic production inspection (see 4.6)

4.3 Sampling.

4.3.1 Preproduction. A preproduction sample of five rate gyros, manufactured in accordance with 3.2, shall be subjected to the preproduction tests detailed in 4.6 at an activity designated by the procuring activity. Acceptance of the preproduction sample shall be based on no defects in the sample, Further production of the rate gyros by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Lot size. Unless otherwise specified in the contract or order, each shipment of units of product shall be a separate "inspection lot" and may differ from the quantity specified in the total order or contract. In any event, all units of a single "inspection lot" shall be made to the same design, with the same materials and manufacturing techniques, and shall be part of one continuous production run.

4.3.3 Periodic production sample. From each succeeding lot or lot of 300 or less that has passed the quality conformance inspection of 4.5, a random sample of five rate gyros shall be selected for submission to an activity designated by the procuring activity to determine compliance of the samples with the requirements of the contract, specification, and drawings.

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The rate gyros shall be considered satisfactory if they meet all specifications and pertinent drawings requirements and withstand the tests specified in 4.5 and 4,6. Failure of the periodic production sample to comply with these requirements will result in the rejection of the lot, as determined by the procuring activity.

4.4 Test equipment. Items of test equipment and power sources listed in Table I are required to perform the tests specified herein. All items shall be of sufficient accuracy and precision to permit valid performance of required tests. Test instruments shall be calibrated at regular intervals and procedures for and results of all calibrations shall be subject to approval of the cognizant Government inspector, Items having equivalent characteristics may be substituted for those identified.

4.5 Quality conformance inspection. Each rate gyro shall be subjected to quality conformance tests and examinations. Failure to meet any requirements specified herein shall be considered cause for rejection. The manufacturer shall furnish and maintain all necessary test-equipment and facilities, and shall provide personnel for performing the quality conformance tests. The test equipment shall be adequate in quantity, and unless definite requirements are specified, shall be of sufficient accuracy and quality to permit performance of the required tests.

4.5.1 Test conditions. Unless otherwise specified, the rate gyros shall be subjected to quality conformance tests under the following conditions:

a.	Temperature:	18°C to 35°C (65°F to 95°F).
b.	Altitude:	Normal ground.
c.	Vibration:	None.
d.	Humidity:	Room ambient to 95 percent relative maximum.

4.5.2 Test procedure for angular rates. Angular rate requirements as specified in 3.3.1 shall be met if the sensitivity (4.5.3) and maximum output (3.3.3) are as specified.

4.5.3 Test procedure for sensitivity.

a. Mount the rate gyro on the gyro fixture (Table I, Item I – Dwg 1874064), using a torque of 20 ± 2 inch pounds on the mounting screws. Position the gyro fixture on the turntable (Table I, Item 2) so that the rate gyro mounting base is at right angles to the plane of the turntable and connector P601 is at the left when viewing the gyro base.

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Table I. - Test Equipment and Power Sources

ITEM	DESCRIPTION	REQUIREMENTS	IDENTIFICATION
1	Gyro Fixture	————	Dwg 1874064
2	Turntable	Horizontal or vertical with continuously adjustable rates, clockwise and counterclockwise, from 0 to 45 degrees per second. Must maintain constant rate within 10 percent of set rate for six minutes. Instantaneous angular velocity shall not vary more than 5 percent from average velocity. Means shall be provided for measuring angular velocity.	————
3	Wattmeter	Two required.	TS-445/U
4	Power Source	115 ± 5.75 volts, 400 ± 8Hz, three phase, 0.5a.	————
5	Power Source	55 ± 1 volt, 400 ± 8Hz, single phase, 30 ma minimum. Total distortion not to exceed 1.0 percent.	————
6	Multimeter	————	AN/PSM-4
7	Resistive Load	25,000 ± 250 ohms.	————
8	Voltmeter	Vacuum tube, ac	Hewlett-Packard Model 400H
9	Megohmmeter	500 volts, capable of measuring 50 megohms.	————
10	Power Source	500 volt, 60Hz	
11	Oscilloscope	————	Tektronic Model 532
12	Camera Attachment	————	Tektronic Model C-12

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b. Using the two-wattmeter method, connect the wattmeters (Table 1, Item 3) between the 115-volt power source (Table I, Item 4) and pins A-B-C of P601 on the rate gyro.

Note: Voltage leads of wattmeters shall be toward the gyro input side.

c. Apply power and take starting power input readings after initial surge subsides (two to three seconds after power application).

d. Correct scale readings obtained in 4.5.3.c by subtracting the power loss due to the potential coils of the respective wattmeters. The power, which is the sum of the corrected wattmeters readings, shall be as specified in 3.5.2.

e. After the gyro has run a minimum of 1-1/2 minutes, note the readings on the wattmeters.

(1) The sum of the wattmeters readings after correction for potential coil power loss shall be as specified in 3.5.3.

(2) The response time for initial power application shall be as specified in 3.3.11.

f. Connect the 55-volt power source (Table I, Item 5) in series with the multimeter (Table I, Item 6) to pins D-E on P601. The current shall be 21 ± 3.15 ma.

g. Connect the resistive load (Table I, Item 7) across pins F-H on P601.

h. Connect the vacuum tube ac voltmeter (Table I, Item 8) across the resistive load.

i. Rotate the turntable in a clockwise direction, when viewed from above, at three speeds such that the voltmeter reads 1 ± 0.75 volts; 3 ± 0.75 volts; and 5 ± 0.75 volts. At each of the three speeds, record the voltage reading and turning rate in degrees per second.

j. Rotate the turntable in a counterclockwise direction, when viewed from above, at three speeds such that the voltmeter reads 1 ± 0.75 volts, 3 ± 0.75 volts and 5 ± 0.75 volts. At each of the three speeds, record the voltage reading and turning rate in degrees per second.

k. Using Figure 1, plot the six points recorded (three points in 4.5.3.i and three points in 4.5.3.j). These points must fall within the limits of the shaded area of Figure 1.

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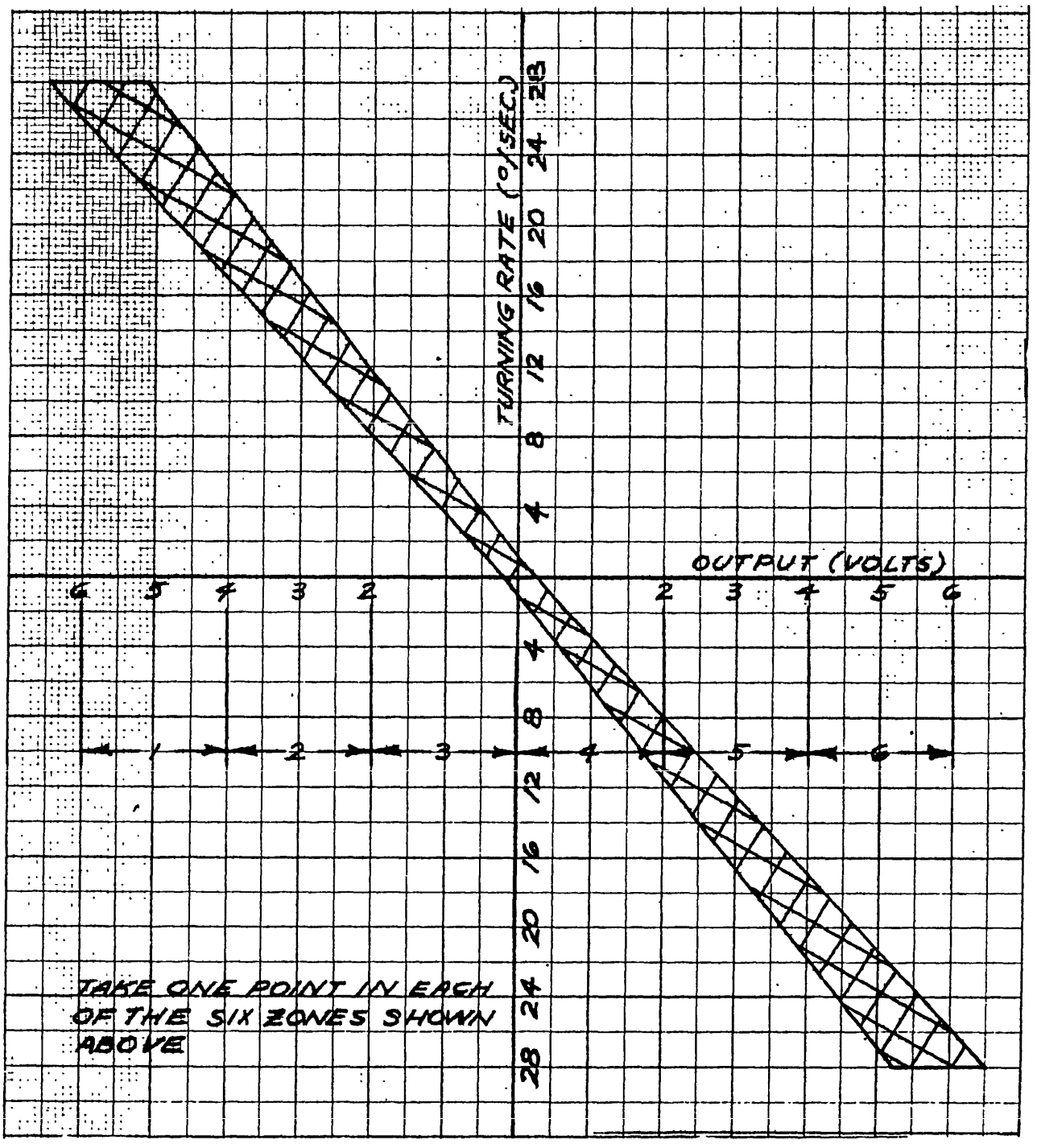


FIGURE 1

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4.5.4 Test procedure for maximum output. To test the rate gym maximum output for the requirements of 3.3.3:

a. Connect the rate gyro and confirm the power and current requirements as described in 4.5.3.b. through g.

b. Remove the cover (see Dwg 1523749).

c. Manually force the gimbal and motor subassembly over to each stopscrew (Dwg 1539007) as shown on Dwg 1503477. The output voltage shall be as specified in 3.3.3, when the stop is against the stopscrew on each side.

4.5.5 Test procedure for phasing. To test the rate gyro phasing for the requirements of 3.3.4:

a. Mount the rate gyro as described in 3.3.4.

b. Connect the rate gyro and confirm the power and current requirements as described in 4.5.3.b. through f.

c. Install a jumper between pins E-F of P601.

d. Set the vacuum tube ac voltmeter on the 100 volt scale and connect it to pins D-H of P601. The voltmeter shall read 55 ± 1 volt if the power source is as specified in Table I, Item 5.

(1) When the rate gyo is rotated in a clockwise direction, (as viewed from above), the voltmeter reading shall increase from 55 ± 1 volt and shall remain at this increased reading as long as the gro is being turned at a steady rate.

(2) When the rate gyro is rotated in a counterclockwise direction (as viewed from above), the voltmeter reading shall decrease from 55 ± 1 volt and shall remain at this decreased reading as long as the gyro is being turned at a steady rate.

4.5.6 Test procedure for zeroing. To test the rate gyro zeroing for the requirements of 3.3.5, mount and connect the rate gyro and confirm the power and current requirements as described in 4.5.3.a. through g.

a. With the rate gyro stationary, the voltmeter shall read 0.15 volt maximum as specified in 3.3.5.1.

b. With the rate gyro rotated at 44 degrees per second minimum clockwise, then stopped, then rotated counterclockwise and stopped, the null voltages shown on the voltmeter shall be within 0.03 volt of the null voltage obtained in 4.5.6.a. This procedure shall be repeated a minimum of three times in each direction.

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c. With the rate gyro resting on either end, or either side, the voltmeter shall read within 0.03 volt of the null voltage obtained in 4.5.6.a. as specified in 3.3.5.2.

4.5.7 Test procedure for spin axis alignment. To test the rate gyro spin axis alignment for the requirements of 3.3.6:

a. Position the gyro fixture on the turntable so that the gyro base is in the same plane as the turntable (with the base down).

b. Connect the rate gyro and confirm the power and current requirements as described in 4.5.3.b. through g.

c. Rotate the turntable either clockwise or counterclockwise, at 10 ± 1 degree per second. The voltage shall not change by more than 0.05 volt from the null voltage obtained in 4.5.6.a.

4.5.8 Test procedure for linearity. If the sensitivity requirements as described in 4.5.3 are met, the linearity as described in 3.3.7 will be obtained.

4.5.9 Test procedure for insulation. To test the rate gyro insulation for the requirements of 3.3.8, the gyro must be at temperatures and humidity conditions specified in 4.5.1 in order to prevent condensation on the gyro parts.

Connect all pins of P601 together. Apply the megohmmeter (Table I, Item 9) across all pins of P601 and ground. The megohmmeter shall read not less than 50 megohms to conform to the requirements of 3.3.8.1.

b. Connect all pins of P601 together. Apply a 500-volt, 60 Hz power source (Table I, Item 10) between all pins of P601 and ground for one minute. There shall be no evidence of insulation breakdown, for compliance with 3.3.8.2.

4.5.10 Visual and dimensional examination. Each rate gyro shall be examined to determine compliance with the requirements for workmanship and dimensions.

4.5.11 Packaging, packing, and marking. Examine the packaging, packing, and marking to determine compliance with the requirements of Section 5.

4.6 Preproduction and periodic production inspection. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspections detailed in 4.5, shall be subjected to the following tests and examinations. When specified in the contract or purchase order, certified inspection data covering the results of preproduction and periodic production inspections shall be forwarded to the procuring activity.

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4.6.1 Test procedure for vibration. The rate gyro, less pack or packaging, shall be mounted securely with the mounting base horizontal to the vibration device by means of its mounting holes and tested as specified in 4.6.1.1 and 4,6.1.2.

4.6,1.1 Vibration, nonoperating. The rate gyro shall be subjected to the "Vibration, nonoperating condition" test of MIL-E-82590 and thereafter shall meet the requirements of 3.6.1.1.

4.6.1.2 Vibration; operating. The rate gyro shall be subjected to the ''Vibration, operating" test of MIL-E-82590 and during and thereafter shall meet the requirements of 3.6.1.2.

4.6.2 Test procedure for high temperature and humidity, nonoperating. The rate gyro shall be subjected to the "High temperature and humidity, nonoperating" test of MIL-E-82590 and thereafter shall meet the requirements of 3.6.2. Condensation from the test chamber shall not be allowed to drop onto the rate gyro. Prior to checkout, wiping or air-hose drying may be performed, but no drying by heating or baking shall be done.

4.6.3 Test procedure for freaqencv. To test the rate gyro frequency for the requirements of 3.3.9:

a. Connect the 115-volt, 400 Hz, three-phase power source to pins A, B, and C of P601. Connect the 55-volt, 400 Hz, single-phase power source to pins D and E of P601.

b. Connect pins F and H of P601 to the input of the oscilloscope (Table I, Item 11).

c. Remove the cover (see Dwg 1523749).

d. Manually force the gimbal and motor subassembly over to a stopscrew (Dwg 1539007) as shown on Dwg 1503477.

e. With the stop against the stopscrew, quickly release the stop.

f. Record the presentation with the camera attachment (Table I, Item 12). From this record, calculate the natural frequency in Hertz (Hz). This frequency must be within the limits specified in 3.3.9.

4.6.4 Test procedure for damping. To test the rate gyro damping for the requirements of 3.3.10, calculate the damping from the record obtained in 4.6.3.f. The damping shall be within the limits specified in 3.3.10.

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4.6.5 Test procedure for operating life, Connect the rate gyro and confirm the power and current requirements as described in 45.3.b. through g. After being subjected to the cycling procedure as set forth in 3.4, the rate gyro shall be capable of performing as specified in 3.3.

Note: Before the start of each cycle set forth in 3.4, the gyro motor shall be allowed to coast to a stop.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 Level A.

5.1.1.1 Cleaning. The rate gyro shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Preservation. None required,

5.1.1.3 Unit packaging. None required.

5.1.2 Level C. Cleaning, preservation, and packaging shall be in accordance with 5.1.1.

5.2 Packing.

5.2.1 Level A. The rate gyro shall be packed in accordance with LD 495855.

5.2.2 Level B. The rate gyro shall be packed in accordance with LD 495855.

5.2.3 Level C. The rate gyro shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with Consolidated Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.3 Marking. Marking of shipping containers shall be in accordance with LD 495855.

6. NOTES

6.1 Intended use. The rate gyro is intended to be used in Torpedo Mk 37 Mods 2 and 3.

6.2 Ordering data. Procurement documents should specify the following:

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a. Title, number, and date of this specification.

b. Any exception to this specification.

c. Required level of preservation and packaging.

d. Required level of packing.

e. Assigned activity for preproduction and periodic production inspections.

f* That additional samples required because of failure of any unit(s) to pass the prescribed periodic production inspections shall be provided and inspected at the expense of the supplier.

6.3 Contract data requirements. Items of deliverable data required by this specification are cited in the following paragraph herein:

Paragraph	Data Requirement	Applicable DIDs
4.6	Certified inspection data	

DIDs (Data Item Descriptions/DD Form 1664) for the above requirements will be documented in the applicable ADL (Authorized Data List). Such data will be delivered as identified on completed (numbered) DIDs when specified on DD Forms 1423 (Contract Data Requirements Lists) and incorporated into applicable contracts.

Custodian:
Navy - OS

Preparing Activity:
Navy - OS
Project No. 1355-N364

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

MIL-G-18853C(OS), Gyro, Rate (For Torpedo Mark 37 Mods 2 and 3)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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Standardization Division (502)
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